

INFO 5651

Fundamental Concepts in Data Science

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Electives	ML Engineer (67.6 / 100)	57.5 / 100
SPECIALIZATION On-Campus Elective		

57

CURRENT AVG SCORE
across 6 job roles

91

POTENTIAL SCORE
with recommended additions

+44

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This intensive course provides a general understanding of the mathematical concepts required for success in data science. This course will cover a wide range of mathematical tools in data science including an overview of calculus and linear algebra along with selected topics from numerical analysis. The course will also explore computational implementations of these ideas. This course provides a bridge for students without these advanced math concepts to learn to apply them within a data science career or within a graduate program in data science.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
ML Engineer	67.6	96.2	+28.6	MEDIUM
AI Engineer	66.9	99.6	+32.7	HIGH
Machine Learning Engineer	64.7	96.4	+31.6	HIGH
Data Scientist	55.6	81.5	+25.9	MEDIUM
Applied Scientist	53.2	97.0	+43.8	HIGH
MLOps Engineer	36.9	76.5	+39.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

✓ Course vocabulary is well-aligned with industry job descriptions.

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

✓ This course is already well-aligned with industry expectations. Continue monitoring for emerging tools and frameworks.

GAP KEYWORDS BY ROLE

ML Engineer	67.6 → 96.2 (+28.6 pts)
learning machine users models model world problems technical	
AI Engineer	66.9 → 99.6 (+32.7 pts)
following models learning software technical model teams platform	
Machine Learning Engineer	64.7 → 96.4 (+31.6 pts)
sets learning models model applications machine technical software	
Data Scientist	55.6 → 81.5 (+25.9 pts)
toolkit sets problems	
Applied Scientist	53.2 → 97.0 (+43.8 pts)
research problems models technical learning industry fundamental world	
MLOps Engineer	36.9 → 76.5 (+39.7 pts)
model scientists learning machine models error	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.